Hasan Ashraf

PhD Student, Mechanical Engineering
University of Alberta
LinkedIn

Email: hashraf3@ualberta.ca
Phone: +1 825-967-9641
Edmonton, Alberta, Canada

Research Summary

Doctoral researcher in Mechanical Engineering specializing in robotics, motion planning, reinforcement learning, and robotic additive manufacturing. Research focuses on constraint-aware motion planning, trajectory optimization, perception, and adaptive multi-robot systems for dynamic manufacturing environments.

Education

University of Alberta, Edmonton, Canada — Jan 2025 – Present
PhD in Mechanical Engineering (Robotics Focus)
Advisor: Dr. Ahmed Jawad Qureshi

National University of Sciences and Technology (NUST), Pakistan — Sep 2019 – Sep 2023
Bachelor of Mechatronics Engineering **(GPA: 3.94/4.0)**

Kennesaw State University, Georgia, USA — Aug 2022 – Dec 2022
Global UGRAD Exchange Scholar **(GPA: 4.0/4.0)**

Publications & Presentations

- **Ashraf, H.**, Irfan, M., Tauqir, M., Radnia, A., Keshmiri, M., Singh, M., Qureshi, A.J. (2026). "Print-forging: A Novel Hybrid Route for High-Performance WAAM." *Proceedings of the 9th Holistic Innovation in Additive Manufacturing Conference (HI-AM 2026)*, Banff, AB, Canada, June 22–23, 2026. **(Under Review; Conference Presentation).**
- **Ashraf, H.**, Naqvi, A.J., Akram, M.U., Khawaja, S.G., Khan, U.S. (2026). "Smart Prosthetics: Vision, Depth Estimation, and Haptics." In: Koucheryavy, Y., Kumar, N., Muthanna, A., Sharma, S. (eds.), *Internet of Things, Smart Spaces, and Next Generation Networks and Systems. ruSMART NEW2AN 2025 2025*. Lecture Notes in Computer Science, vol. 16462. Springer, Cham. `https://doi.org/10.1007/978-3-032-19981-2_20`. **(Published; Conference Presentation).**
- **Ashraf, H.**, Singh, S., Qureshi, A.J. (2025). "Optimizing Process Parameters for VANCAR-O Deposition in Wire Arc Additive Manufacturing." *CSME-CFDSC-CSR 2025 International Congress*, Montreal, QC, Canada, May 25–28, 2025. `https://doi.org/10.82417/t4hq-k109`.**(Conference Presentation).**

Research Experience

PhD Researcher — Jan 2025 – Present
ADaMS Lab, Dept of Mechanical Engineering, University of Alberta — Edmonton, AB

- **Spatio-Temporal Motion Planning:** Developing constraint-aware motion and trajectory planning methods for collaborative multi-robot systems, addressing collision avoidance, temporal constraints, and dynamic workspace changes in manufacturing environments.
- **AI & Reinforcement Learning:** Investigating sampling-based planning, trajectory optimization, and deep reinforcement learning for adaptive and online robot motion planning under process and environmental constraints.
- **Robotic Additive Manufacturing:** Conducting research on Wire Arc Additive Manufacturing (WAAM), including VANCAR-O deposition, process parameter optimization, and integration of robotic systems for adaptive manufacturing.
- **Simulation & Robotics:** Developing and evaluating robotic algorithms using **ROS 2, NVIDIA Isaac Sim, NVIDIA Isaac Lab, and Python**, with emphasis on GPU-accelerated simulation and reinforcement learning.

Mitacs Research Graduate Intern — Sep 2025 – Present
Elementiam & ADaMS Lab, University of Alberta — Edmonton, AB

- **Constraint-Aware Motion Planning:** Developing optimization-based trajectories for 6-DOF manipulators on $SE(3)$ manifolds, ensuring strict adherence to torch-angle and standoff distance constraints during robotic welding.

- **Perception-in-the-loop:** Architecting a pipeline to transform noisy 3D point cloud data into Euclidean Signed Distance Fields (ESDFs) for real-time collision avoidance in dynamic environments.
- **Algorithmic Evaluation:** Comparative analysis of **CHOMP** and **TrajOpt** frameworks for feasibility in hybrid manufacturing, focusing on singularity avoidance and joint-space continuity.
- **Autonomous Scanning:** Investigating "Next-Best-View" (NBV) planning strategies for surface reconstruction of unknown workpieces to minimize occlusions in robotic inspection.

Projects

Articulated4D: Generative Modeling of Articulated Assets 2026

- Investigated the use of **State-Flow Matching** to synthesize 4D temporal sequences from static 2D images.
- Implemented a latent diffusion-based training pipeline on NVIDIA H100 clusters, focusing on joint-limit consistency in generative meshes.

Vehicle State Estimation using GPS-IMU sensor fusion 2025

- Developed a robust **Kalman Filter (KF)** for vehicle localization, incorporating a χ^2 innovation gate to mitigate the impact of GPS interference.
- Achieved a **1.9m RMSE** localization accuracy on the KITTI dataset.

Anomaly Detection via Reverse Distillation with Balanced Few-Shot Mask Supervision 2025

- Extended RD4AD with pixel-wise few-shot supervision and mask-aligned feature loss to detect medical parasites under severe foreground-background imbalance.
- Introduced a push–pull feature learning strategy, improving anomaly localization by up to **25%** Pixel-AUPRC over baseline few-shot methods.

Technical Skills

Robotics & Motion Planning: Path Planning (RRT, RRT*, PRM, PRM*, OMPL), Trajectory Optimization, Model Predictive Control (MPC), Manipulator Kinematics & Dynamics, Collision Detection & Avoidance, Multi-Robot Motion Planning.
Reinforcement Learning & AI: Deep Reinforcement Learning (DRL), Markov Decision Processes (MDPs), Policy Optimization, Imitation Learning, Generative Models, Computer Vision, Sensor Fusion.
Programming: Python, C++, CUDA, Linux, Git.
Robotics Frameworks & Simulation: ROS, NVIDIA Isaac Sim, NVIDIA Isaac Lab, Gazebo, PyBullet, RViz, RoboDK.
Libraries & Tools: Solidworks, MoveIt/MoveIt 2, OMPL, SKRL, Open3D, URDF/USD, PyTorch, NumPy, SciPy, OpenCV.

Scholarships and Awards

- **Graduate Recruitment Scholarship**, University of Alberta 2025
- **Global UGRAD Fellowship**, U.S. Department of State (Highly Competitive) 2022
- **NUST High Achiever Award** (Top 1% of Engineering Cohort) 2019 – 2023